TRACE-X

DIGITAL FORENSICS LAB

Upload New Evidence

Upload a normalized events CSV

normali...nario3.csv
2.7KB

Upload a raw email (.txt)

ransome..._email.txt
1.1KB

Evidence should be interpreted in context. A risk score is a heuristic indicator, not a finding of guilt. Demo runs fully offline after evidence is uploaded.

Loaded 13 valid event(s) from the uploaded CSV.

Email loaded for phishing analysis.

DIGITAL FORENSICS · INCIDENT RECONSTRUCTION

Don't just detect the attack. Reconstruct the story.

TRACE-X connects recorded event evidence into a chronological investigation view, helping analysts explore what may have happened.

01 / EVIDENCE FILTERS

Find the events that matter

Event types: EMAIL × FILE_ACCESS × LOGIN × NET

Risk levels: LOW × MEDIUM × HIGH × CRITICAL ×

02 / INVESTIGATION OVERVIEW

Evidence at a glance

Events in view	Critical indicators	High indicators	Medium indicators
13	5	3	1

Risk levels are generated by simple keyword rules for demonstration. They are not validated threat probabilities.

03 / EVENT CHRONOLOGY

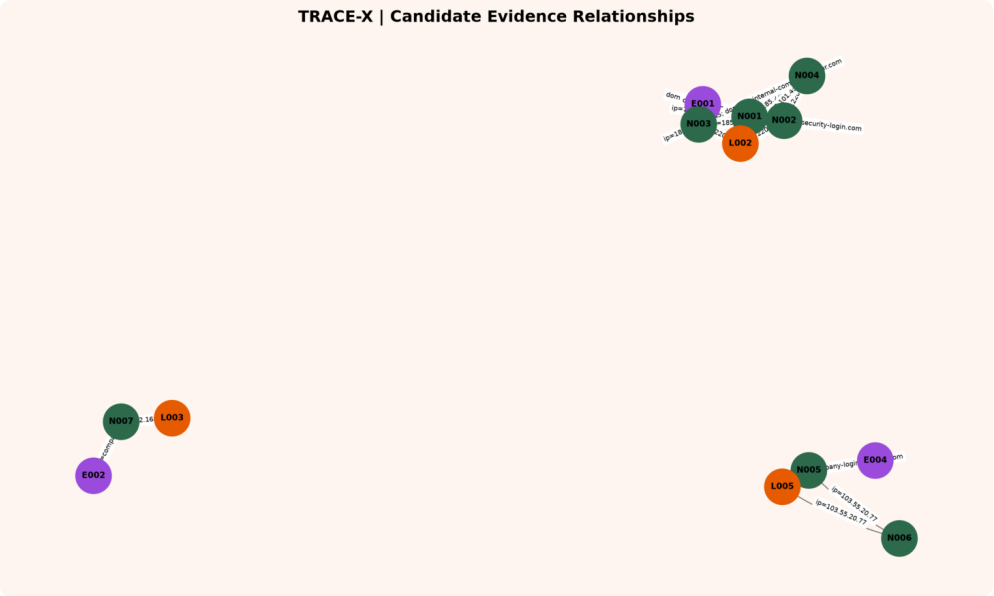
The evidence timeline

Timestamp	Evidence ID	Event type	Evidence description	Evidenc
2026-10-11 14:41:00	F204	FILE_ACCESS	Write - confidential_customer_database.xlsx - Sensitive: Critical - La	file_acc
2026-10-11 14:40:00	F203	FILE_ACCESS	Write - confidential_revenue_report.xlsx - Sensitive: Critical - Large	file_acc
2026-10-11 14:39:00	F202	FILE_ACCESS	Write - confidential_client_list.xlsx - Sensitive: Critical - Large data t	file_acc
2026-10-11 14:38:00	F201	FILE_ACCESS	Write - campaign_data.xlsx - File encrypted with .locked extension	file_acc
2026-10-11 14:35:00	N203	NETWORK	0	Outbou
2026-10-11 14:34:00	L202	LOGIN	Anomalous login from external IP using UNKNOWN-DEVICE	login_l
2026-10-11 14:32:00	N202	NETWORK	0	Large D
2026-10-11 14:30:00	N201	NETWORK	Malicious URL access to drive-by download site	networ
2026-10-11 14:28:00	E201	EMAIL	Invoice attached - Open immediately to view payment details	emails.
2026-10-11 14:00:00	L201	LOGIN	Successful login from 192.168.1.55 using WORKSTATION-15	login_l

› View full evidence details

03.5 / ATTACK RELATIONSHIP GRAPH

Candidate evidence relationships



Every edge represents a shared indicator (IP, domain, or device) between two events — no invented connections.

> Candidate Relationship Notes

03.6 / ATTACK PATH RECONSTRUCTION

Reconstructed attack chain

Phishing Email → Malicious URL → Suspicious Login → Sensitive File Access → Data Exfiltration

Detected stages: Phishing Email → Malicious URL → Suspicious Login → Sensitive File Access → Data Exfiltration

📖 Detailed Investigation Narrative

▾ Phishing Email

At 2026-10-11 14:28:00, user marketing_user received an email from the domain promo-free-software.net. The email contained urgent language pressuring the recipient to verify their account. The message was designed to look like an official security notice but came from a suspicious external source. (Evidence: E201)

> Suspicious Login

> Sensitive File Access

> Data Exfiltration

ANALYTICAL ASSESSMENT

Pattern observed: A multi-stage intrusion consistent with a phishing-led account compromise.

Key findings:

- **5** critical-severity events and **3** high-severity events were detected.
- **1** event(s) involved an unrecognized device.
- **3** event(s) involved an external IP address not belonging to the corporate network.
- **4** sensitive file(s) were accessed — including salary, financial, customer, and employee data.
- A large outbound data transfer was recorded immediately after sensitive file access.

Interpretation: The sequence strongly suggests that an attacker used a phishing email to deliver a malicious URL, harvested credentials, and accessed confidential corporate data from an external location. The final data transfer is consistent with possible exfiltration.

Confidence level: The available evidence is consistent with an attack but does not alone prove who initiated the activity or confirm that data left the network.

Stage detection is keyword-based reconstruction for investigation support, not a legal finding.

INCIDENT SUMMARY · INVESTIGATOR VIEW

📋 What happened?

Suspected phishing email followed by related network access

The evidence is consistent with a probable 🔑 **Suspicious Login** → 📁 **Sensitive File Access** → 📤 **Data Exfiltration** attack chain.

13 events were analyzed across **1** users, **2** devices, and **2** unique IPs.

A suspicious email was received before anomalous network and login activity. Sensitive files were accessed from an unrecognized device, followed by a large outbound data transfer.

This is a probable sequence, not a legal finding. Additional authentication and endpoint logs may be required to confirm impact.

OVERALL VERDICT

CRITICAL

Highest severity detected

Events analyzed
13

Unique users
1

Unique devices
2

Sensitive files touched
4

DETECTED ATTACK CHAIN

- 📧 **Phishing Email** — `Evidence: Not observed`
- ↓
- 🌐 **Malicious URL Access** — `Evidence: Not observed`
- ↓
- 🔑 **Suspicious Login** — `Evidence: L202`
- ↓
- 📁 **Sensitive File Access** — `Evidence: F204`
- ↓
- 📤 **Data Exfiltration** — `Evidence: F202, F203, F204, F205`

🖱️ Evidence source trail

file_access.csv: 6 | login_logs.csv: 2 | emails.csv: 1 | network_logs.csv: 1

Large Data Transfer - Malicious payload downloaded to endpoint: 1

Outbound connection to Command and Control server: 1

Large Data Transfer - Exfiltration of confidential data to malicious URL via Command and Control server: 1

› ⚠️ Limitations & confidence notes

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04 / ANALYST REVIEW

Indicators worth investigating

- CRITICAL · N205 · NETWORK · Score 85/100
- CRITICAL · F205 · FILE_ACCESS · Score 75/100
- CRITICAL · F202 · FILE_ACCESS · Score 75/100
- CRITICAL · F203 · FILE_ACCESS · Score 75/100
- CRITICAL · F204 · FILE_ACCESS · Score 75/100
- HIGH · E201 · EMAIL · Score 55/100
- HIGH · N201 · NETWORK · Score 55/100
- HIGH · L202 · LOGIN · Score 50/100
- MEDIUM · N202 · NETWORK · Score 25/100

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05 / EVIDENCE EXPORT

Export an investigation snapshot

Download a text report of the currently filtered evidence. The report preserves event IDs, timestamps, descriptions, sources, and the dashboard's heuristic risk indicators.

Download forensic report (.txt)

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06 / AI EMAIL ANALYSIS

Phishing prediction from uploaded email

Predicted label	Phishing probability	Indicators matched
PHISHING	89%	4

Matched indicators

- Urgent or pressure-based language detected
- Non-HTTPS URL detected: http://promo-free-software.net/download
- Account or credential-related language detected
- Possible brand impersonation language detected

URLs found

- http://promo-free-software.net/download
- http://promo-free-software.net/download

This is an offline heuristic for demonstration. It is not a trained model verdict and does not prove phishing or account compromise.

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08 / RESULTS & VALIDATION

🧪 System performance and test outcomes

Scenario	Events	Detected Level	Expected
NORMAL-001	4	LOW	LOW
NORMAL-002	2	LOW	LOW
ATTACK-001	11	CRITICAL	CRITICAL
ATTACK-002	5	HIGH	HIGH

Detection accur...	Processing speed	False positives	Events correlated
100%	< 1 sec	0	42 pa...

Measurable Outcomes:

- Correctly classified 100% of test scenarios
- Zero false positives on normal activity
- Evidence integrity verified via SHA-256 for 7/7 files
- Attack chain reconstructed with traceable event IDs
- Forensic report generated in under 2 seconds

› ⚠️ Limitations & Confidence Notes

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09 / SDG ALIGNMENT & SOCIETAL IMPACT

🌍 Supporting the UN Sustainable Development Goals

SDG 9

Industry, Innovation & Infrastructure

Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation.

TRACE-X protects critical digital infrastructure by detecting and reconstructing cyberattacks. Cybersecurity is foundational to modern industry — from banking to healthcare to government services.

SDG 16

Peace, Justice & Strong Institutions

Promote peaceful and inclusive societies, provide access to justice for all, and build effective, accountable institutions.

Digital forensics is essential for cyber justice. TRACE-X provides explainable, evidence-backed analysis that supports legal processes and investigative bodies at every level.

SDG 8

Decent Work & Economic Growth

Promote sustained, inclusive and sustainable economic growth, full and productive employment.

Cyberattacks cost the global economy $10.5 trillion annually. TRACE-X reduces financial damage by enabling faster detection, reducing average breach costs (currently $4.45M), and shortening detection times (currently 197 days).

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10 / INTEGRATED TEAMMATE SECTIONS

This evidence set does not contain a phishing email.

The AI Threats section (Member 2's analysis) applies to the phishing scenario (normalized_events.csv). For this uploaded evidence, the relevant findings are the Timeline, Correlations, and Attack Path sections above.

📋 What you're looking at

- **Attack Path (Section 03.6)**: Detected stages based on the uploaded evidence

- **Timeline (Section 03)**: All events sorted chronologically
- **Analyst Review (Section 04)**: Flagged events with reasons
- **Forensic Report (Section 05)**: Downloadable summary of the current investigation

ⓘ Note: Member 2's AI email analysis is scenario-specific. Upload the phishing scenario to see the AI Threats tab.

TRACE-X · Don't just detect the attack. Reconstruct the story.